ETL MODERNIZATION ON GOOGLE CLOUD

Legacy ETL → Google Cloud · ODI first

Measure the estate in 3–4 weeks, then migrate in waves at a price set by the inventory

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Four phases converted in waves; what is needed to start. Appendix from p. 9. p. 7 – 8

Why Now — What a Legacy ETL Estate Usually Looks Like

KOMTAS

Findings that come up again and again when an ODI estate is inventoried, and what changes once it runs on Google Cloud.

01 · WHY NOW

Production only

No separate development or test environment; every change is made live.

No automatic alerting

Failures are found by a night operator or by users in the morning.

Knowledge with a few people

Often the vendor that built it; the owner cannot see all the logic.

No data quality checks

Duplicates, nulls and reconciliation are not checked systematically.

Hundreds of small jobs

Most jobs run in under a minute; the complexity is in the count.

Licence and skills cost

Oracle-based ETL and DWH licences; ODI skills are getting scarce.

Each of these is checked in the assessment and becomes a line in the migration plan.

WHAT CHANGES ON GOOGLE CLOUD

Dev, test and prod
separated, deployed by CI/CD

An alert on every failure
Cloud Monitoring, before users notice

Transformations as SQL
versioned, tested, with lineage (Dataform)

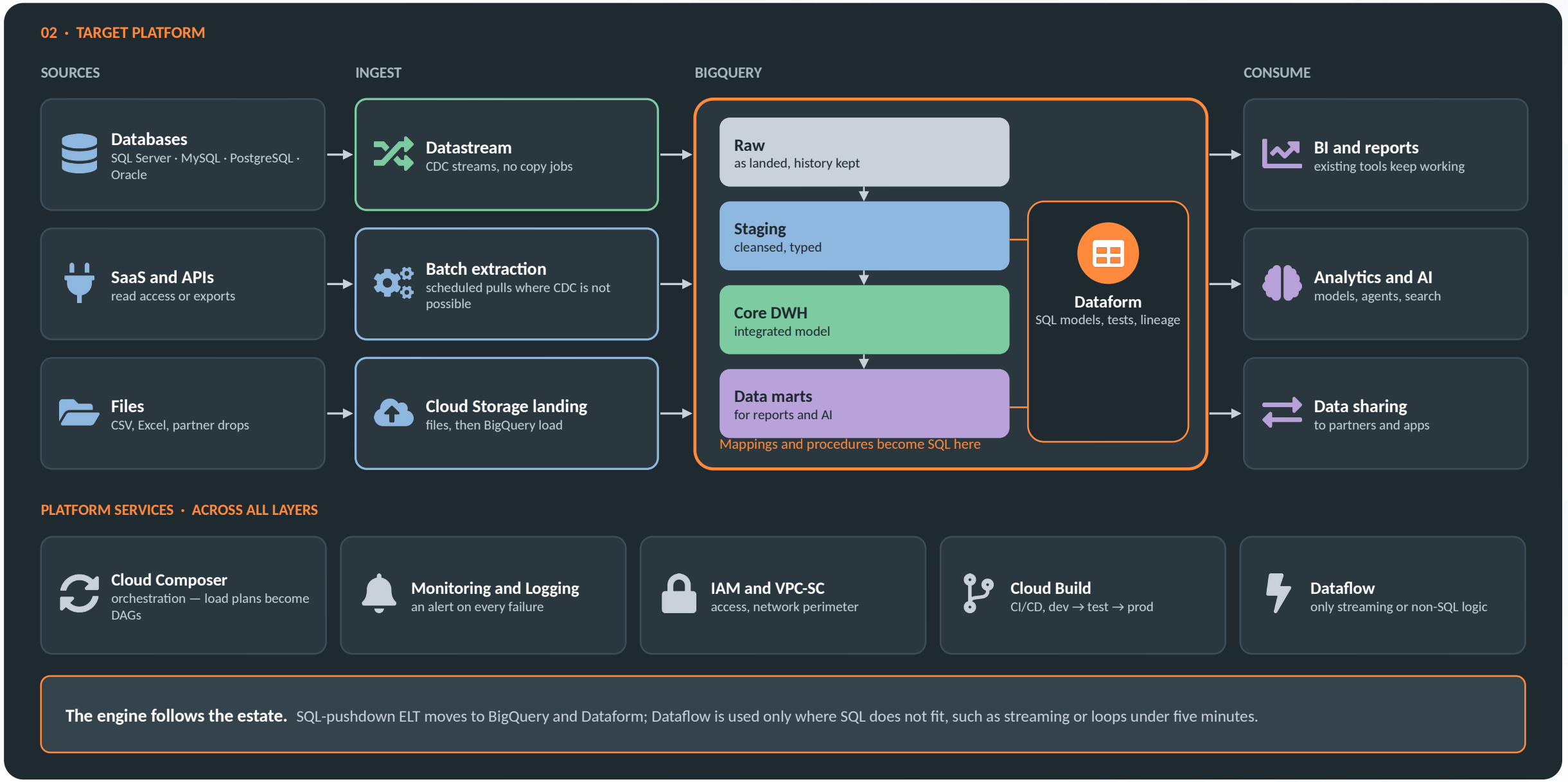
Ingestion as configuration
Datastream CDC instead of copy jobs

Pay for use
serverless BigQuery, no ETL server

Target Platform on Google Cloud

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Sources land through CDC, batch extraction or files; transformations run as SQL in BigQuery; one orchestrator, one monitoring layer.



How ODI Maps to Google Cloud

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Every ODI object type has a target pattern. Each pattern is built once in the pilot and reused for every package of that type.

02 · PATTERN CATALOGUE

ODI OBJECT	GOOGLE CLOUD	HOW IT MOVES	
Mapping / Interface	Dataform SQLX model	SQL translated with BigQuery Migration Service, then reviewed	translate + review
Procedure (PL/SQL)	BigQuery scripting or stored procedure	logic rewritten, tested against the old output	rewrite
Package	Composer task group	steps and branches kept in the same order	template
Load Plan	Composer DAG	serial and parallel steps, retries, schedule	generated
Variable	Airflow or Dataform variable	refresh queries moved to the first task	template
Journalizing (CDC)	Datastream stream	no code; source CDC must be enabled	configuration
File LKM	Cloud Storage + BigQuery load	one landing pattern for all files	template
Knowledge Module	Generic landing and merge pattern	written once, reused by every package	template
Operator log	Cloud Logging + Monitoring	alerts, run history, dashboards	configuration

configuration lowest effort

template low

translate + review medium

rewrite highest

effort per package

Two Services — Measure First, Then Migrate

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The assessment turns an unknown estate into an inventory and a price. The migration is then run in waves against that inventory.

03 · TWO SERVICES

A — ETL Migration Assessment

Fixed price · 3–4 weeks

Focus	Measure the estate and price the migration
Evidence	Repository queries, discovery form, interviews
Access	Read-only to the ODI repository, or the script run by your team
Output	Inventory, complexity groups, target architecture, plan and price

B — ETL Migration on Google Cloud

In waves · duration set by the estate

Focus	Build the platform, convert, run in parallel, cut over
Pricing	Per package by complexity group, or time and materials with a cap
Team	Architect, data engineers, cloud engineer, project manager
Output	Running platform on Google Cloud, ODI switched off, runbooks handed over

Why not a fixed price straight away?

When an estate was built by a third party and is not fully documented, a fixed price either carries a large risk margin or breaks at the first surprise. The assessment removes that uncertainty first; the migration price then follows the measured inventory.

How the Estimate Is Built

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From the repository to a price in six steps. Unit hours are assumptions until the pilot calibrates them.

03 · ESTIMATE

1 Discovery script
reads the ODI repository: packages, mappings, procedures, load plans, runtime

2 Inventory
every package with sources, targets, schedule, run time and volume

3 Complexity groups
five structural groups, plus 1:1 ingestion handled as configuration

4 Unit hours × reserve
analysis, conversion and test hours per group; reserve for unknowns

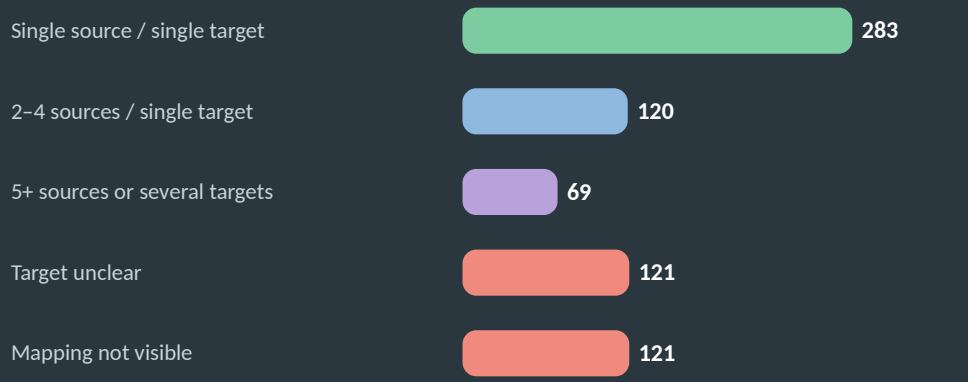
5 Pilot calibration
two critical flows end to end; unit hours corrected

6 Plan and price
team size against target date; per-package or capped T&M price

A recent retail estate

example · anonymised

714	15	236	daily
packages	source systems	source tables	all flows



242 packages (34%) need a closer look — usually procedure-heavy. The deeper discovery queries resolve them before the price is fixed.

About 290 jobs are 1:1 copies. If CDC can be enabled on the sources, they become Datastream configuration instead of code.

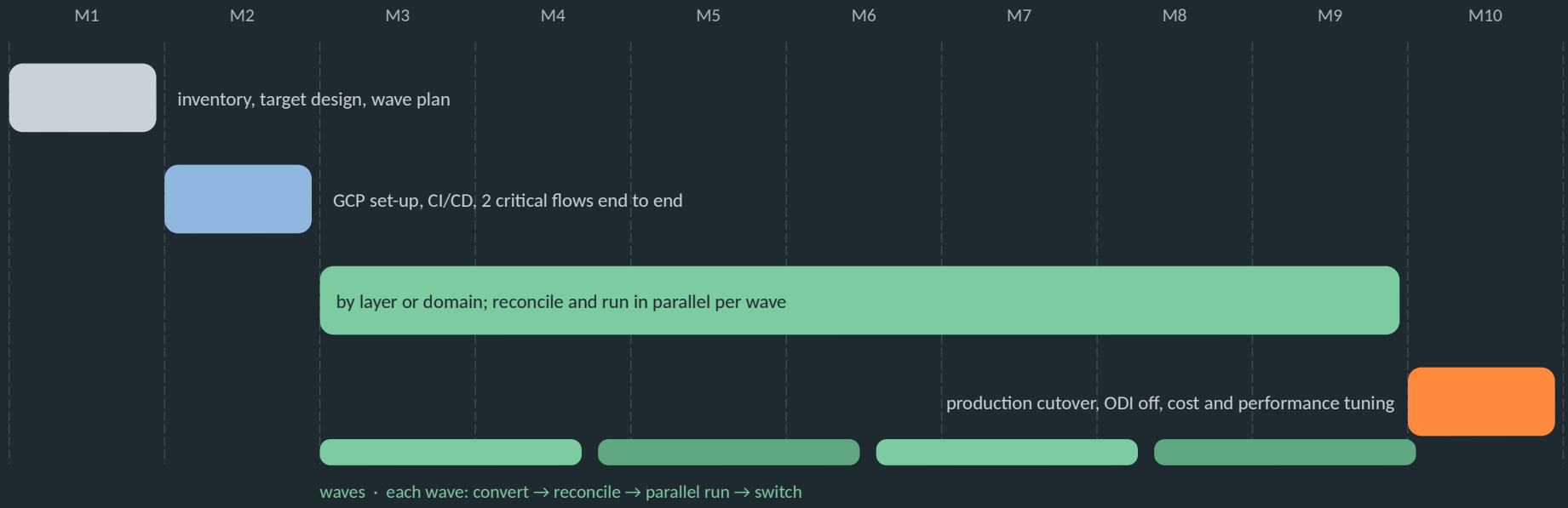
Migration Plan — Four Phases, Converted in Waves

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Phase 3 length is set by the inventory and the team size. Every wave is reconciled and run in parallel before it is switched over.

04 · MIGRATION PLAN

PHASES · INDICATIVE MONTHS



TEAM

- **Cloud Solution Architect** full in phases 1–2, about 20% after
- **Data Engineers** 2 to 5, set by the target date
- **Cloud Engineer / DevOps** full time
- **Project Manager** part time

QUALITY GATES

- **Reconciliation** row counts and checksums per table
- **Parallel run** old and new side by side
- **Cutover runbook** steps, owners, rollback

What We Need From You — and the Next Steps

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Three things make the assessment possible; the first answer on CDC already changes the size of the migration.

04 · NEXT STEPS

WHAT WE NEED

Repository access

day one

Read-only access to the ODI work repository, or your team runs the discovery script. About one day.

A CDC answer per source

first week

Can change data capture be enabled on each source database? SaaS sources may need batch extraction.

The people who know

workshops

Platform owner, the team or vendor that built the flows, owners of the critical reports.

TWO THINGS THAT SET THE DATE

Access approvals

Repository read access, and source access for CDC tests. Each is a separate approval.

Raised as one list in week 1

Run history kept

How many days of session history ODI keeps. Runtime and volumes come from it.

Under 30 days narrows the view

NEXT STEPS

1. **Walkthrough** — today
   you are here
2. **Discovery script run** — about one day
3. **Assessment proposal** — fixed price, 3–4 weeks
4. **Kick-off** — inventory, CDC check, design

Measure the estate first. The price follows the inventory, not the other way round — and the inventory stays with you even if the migration is run differently.

APPENDIX

Detail — for reference and questions

A	**Discovery and effort** Discovery toolkit · Effort model	p. 10 – 11
B	**Accelerators** Google Cloud services and funding programmes	p. 12
C	**Delivery and scope** Reconciliation, parallel run and cutover · Scope and limits	p. 13 – 14

Discovery Toolkit — Form and Repository Queries

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The form collects what only people know; the queries read everything the repository already holds. Nothing is written to the repository.

DISCOVERY FORM · 13 SHEETS

- 0 Instructions
- 1 System architecture
- 2 Data inventory
- **3 Pipeline and job inventory ★**
- **4 ETL type split ★**
- **5 Performance and SLA ★**
- 6 Dependencies
- 7 Data quality
- 8 Error handling
- 9 Access and repository
- 10 Risks
- 11 Expected outputs
- 12 Sample flow

★ pre-filled by the queries; the customer adds business meaning only

REPOSITORY QUERIES · ODI 12c, READ-ONLY

#	QUERY	WHAT IT READS	FEEDS
00	Schema check	columns of every repository table used, before anything runs	—
01	Package inventory	steps, mappings, procedures, sources, targets, load plan	Sheet 3
02	Effort group	the five structural groups and 1:1 copy candidates	Effort model
03	Mapping complexity	joins, filters, expressions, aggregates, KMs	Unit hours
04	Procedures	lines, PL/SQL, dynamic SQL, DB links	Blind spot
05–07	Variables, CDC, files	refresh queries, journalizing, non-DB sources	Patterns
08	Load plans	structure, parallel steps, schedules	DAG design
09	Runtime	runs, min/avg/max minutes, success rate, errors	Sheet 5
10–11	Volumes	rows per day, table sizes on the target DWH	Sheet 2
12	Summary	counts per prefix, group and technology	Proposal

Effort Model — Groups, Unit Hours and Reserve

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Package counts come from the inventory; unit hours are starting values that the pilot corrects. Person-days = hours ÷ 8 × (1 + reserve).

UNIT HOURS PER PACKAGE · STARTING VALUES

EFFORT GROUP	ANALYSIS	CONVERSION	TEST	TOTAL
Single source / single target	0.5	2	0.5	3
2–4 sources / single target	1	3	1.5	5.5
5+ sources or several targets	2	4	2	8
Target unclear	1	3	1.5	5.5
Mapping not visible	1	3	1.5	5.5
1:1 ingestion — configuration	0.1	0.5	0.25	0.85

Grouping order: mapping not visible → target unclear → 5+ sources or several targets → 2–4 sources → single source. The 1:1 ingestion group applies only when CDC is possible on the source.

PHASES

PHASE	DAYS	WHO
Analysis and design	20	1 DE, cloud engineer, architect
Platform and pilot	20	2 DE, cloud engineer, architect
Conversion	effort ÷ DE	all DE; architect 20%
Cutover	20	2 DE, cloud engineer

Example · 714 packages

Base effort	**424 pd**
Reserve 20%	**84.8 pd**
Total	**508.8 pd**
3 DE	**≈ 11.5 months**
5 DE	**≈ 9 months**

If the ~290 copy jobs become Datastream configuration, the total falls by roughly 90 person-days — about one month with 3 data engineers.

The model ships as a workbook: paste the inventory, set the team size, read the plan and the Gantt.

Accelerators and Funding

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Google Cloud services and programmes that shorten the migration or lower its cost. Eligibility and terms are set by Google per project.

BigQuery Migration Service	Batch SQL translation from Oracle and other dialects into BigQuery SQL; used for mapping and procedure SQL. The Migration API is offered at no charge.
Datastream	Serverless change data capture from MySQL, PostgreSQL, Oracle and SQL Server into BigQuery — replaces 1:1 copy jobs.
Dataform	SQL workflows in BigQuery with dependencies, assertions (tests), documentation and lineage; versioned in Git.
Cloud Composer	Managed Apache Airflow; ODI load plans and packages become DAGs and task groups.
Google Cloud RAMP	Partner-led migration programme with assessment support and funding or credits, subject to Google's eligibility review.

Product capabilities change; the assessment confirms current support for the exact source versions.

Reconciliation, Parallel Run and Cutover

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How each wave proves the new platform returns the same numbers before the old flow is switched off.

1 Reconcile

Every converted table is compared with its ODI counterpart after each load.

- Row counts per table and per load date
- Checksums on key and amount columns
- Aggregates on the measures reports use
- Differences logged with an owner and a cause

2 Run in parallel

Old and new run side by side on the same days.

- Typically two to four weeks per wave
- Daily reconciliation report to both teams
- Business owners confirm the critical reports
- Exit only when differences are explained

3 Cut over

A written runbook per wave, rehearsed once.

- Freeze, final load, switch of consumers
- Named owner for every step
- Rollback path kept until sign-off
- ODI flow disabled, then removed

Acceptance is a number, not an opinion. Each wave is accepted when reconciliation is within the agreed tolerance for the agreed period, and the business owner has signed the critical reports.

Scope and Limits

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What the services cover, what they leave out, and where they cannot be offered today.

IN SCOPE

- ODI packages, mappings, procedures, variables and load plans
- Ingestion from the existing sources into BigQuery
- Target model layers as they exist today (raw to marts)
- Orchestration, monitoring, CI/CD and access set-up
- Reconciliation, parallel run, cutover, ODI decommissioning
- Runbooks and handover to the operating team

OUT OF SCOPE

- Changes to the source systems themselves
- Redesign of the data model beyond agreed fixes
- Migration of BI reports and dashboards (separate scope)
- Licence costs of Google Cloud consumption
- Data that must stay on premises or in Türkiye by regulation

Regulated workloads

Workloads that must stay in Türkiye under BDDK or similar rules cannot move until a Google Cloud region in Türkiye is open. A region was announced with Turkcell in November 2025; its opening date is not yet set. Until then the services address retail, consumer, e-commerce, media, manufacturing and holding companies.